
Plan Overview

A Data Management Plan created using DMPonline

Title: A Safe Bet: design and pilot evaluation of a player-tailored online responsible gambling promotion framework

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Template: EUR Data Management Plan v4.4

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Project abstract:

The worldwide surge in online gambling opportunities has led to increasing demands to upscale national gambling regulations in order to promote (online) responsible gambling (RG) behaviors. However, scientific evidence on currently available online RG tools is limited, online RG tools are often focused only on players at higher risk, overlook key predictors of escalation into dysfunctional patterns of gambling behavior, and are poorly targeted and timed. Moreover, the limited collaboration between the gambling industry and the scientific research field has prevented the effective translation of scientific knowledge into evidence-based RG applications. This project is the result of a collaboration between academia and the gambling industry, with the goal of combining efforts to design online RG solutions more effectively protecting gamblers across all level of risk. Our goal is to develop and evaluate a comprehensive preventative intervention framework for online RG delivering high-quality, evidence- and theory-based RG tools to players based on their individual level of risk and pattern of gambling behavior. Our innovative RG framework combines a gambling risk assessment algorithm tracking players' behavior on a gambling website, with a newly designed library of RG micro-interventions to scaffold a player-tailored RG promotion model. The project will run over 2 years encompassing a) the iterative development of the standardized library of RG tools based on the synthesis of existing scientific evidence and theory on RG interventions; b) the development of a recommendation system connecting the library of interventions with the risk assessment algorithm by using the power of artificial intelligence systems; c) the evaluation of a proof-of-principle (POC) version of the system, which will inform the development of the full RG framework.

The POC version of the RG framework will be introduced to customers of two gambling operator websites (in NL and the UK), deploying RG interventions mandated by gambling

regulations and a selection of RG tools holding scientific promise. The evaluation of the POC will focus on feasibility and user acceptance (i.e., to what extent players engage with the RG tools, what is their experience with them), and potential impact on gambling risk in a mixed-methods study collecting quantitative and qualitative data. All players will dynamically receive sets of RG tools tailored to their current level of risk over a period of 6 months, with their level of risk assessed daily. A subgroup of players will be further interviewed on their experience with the tools at conclusion of the quantitative data collection. The results will inform the final development of the full RG framework, including the entire library of interventions with a more advanced, self-learning version of the recommendation system, which will be able to decide which RG tools are best suited for each player to minimize escalation of risk, while not interfering with their play experience. With this project, we hope to support positive play via safety promotion for all and provide targeted interventions for harm reduction at the higher levels of gambling risk.

ID: 104547

Start date: 01-10-2022

End date: 01-10-2024

Last modified: 25-04-2024

A Safe Bet: design and pilot evaluation of a player-tailored online responsible gambling promotion framework

General

Please tick the following boxes if you agree to act according to the following terms:

- I will answer all questions truthfully and to the best of my knowledge
- I will discuss the data management plan with my research team
- I will check and, if necessary, update my data management plan a minimum of once a year

Support in writing a data management plan is available through the faculty Data Stewards. If your faculty has no Data Steward, support is available through the Research Data Management Specialist of the Erasmus Data Service Centre (EDSC is part of the University Library). Which research support professional is available for you?

- Data Steward of my own faculty - ESSB

Scientific research must be conducted in line with existing guidelines on good research practices and integrity. Please tick the boxes if you have read and understand these guidelines and will act accordingly.

- The European Code of Conduct for Research Integrity (ALLEA, 2017)
- The Netherlands Code of Conduct for Research Integrity (VSNU, 2018)

Administration & Project Description

Project title

A Safe Bet: development and pilot evaluation of a player-tailored online responsible gambling promotion framework.

Project start date as intended

2022-10-01

Project duration in months as intended

36

Funding body (if applicable)

Project (partially) funded by collaborating gambling operators and Large Grant from the International Centre for Responsible Gambling (ICRG)

Grant number (if applicable)

Not applicable.

Date of DMP Version 1

2022-08-01

Current DMP - Version [if other than version 1]

25/04/2024

Current DMP - Date [if other than version 1]

Question not answered.

List the name and affiliation of all members of the research team.
List the researcher responsible for research data management first.
For PhD projects, please indicate the Promotor(s) and/or Daily Supervisor(s) with a (!)

	Name	Email	ORCID	Research Institution	Role
1	Harriet Galvin	galvin@essb.eur.nl	0000-0003-0698-5093	EUR	PhD candidate
2	Marilisa Boffo, Phd	boffo@essb.eur.nl	0000-0003-4730-7838	EUR	Daily Supervisor / Principal Investigator
3	Reinout Wiers, Phd	r.wiers@uva.nl	0000-0002-4312-9766	UvA	Co-promotor / Co-principal Investigator
4	Leroy Snippe	l.snippe@uva.nl	NA	UvA	Consultant
5	Rutger Engels, Phd	rutger.engels@essb.eur.nl	0000-0003-1944-9126	EUR	Main promotor

Briefly summarize the project background and research question(s) to help others understand the purpose for which the data are being collected or created.

The overarching goal of this collaborative project is to design, develop and evaluate a preventative intervention framework for online responsible gambling (RG) in a naturalistic setting. The envisioned prototype of the online RG intervention framework combines a validated online gambling risk assessment algorithm using behavioural tracking data with a newly designed, evidence- and theory-based library of micro-interventions and digital nudges to scaffold a player-tailored RG promotion model. This project is Phase I of a larger research collocation endeavour and will look to evaluate a first proof-of-concept (POC) version of the RG framework. We intend to leverage the unique opportunity of the online gambling infrastructure, utilizing web-based design mechanics, data science, and behavioural science principles. The comprehensive system approach proposed here will a) use player behavioural tracking data to identify gambling risk status based on a validated machine learning algorithm modelling complex, non-linear relationships between multiple predictors to explicitly account for variability in gambling risk markers; and b) engage in player safety promotion for all players as well as targeted interventions aimed at harm reduction in at-risk gamblers. Our aim is to investigate an evidence-based, data-driven, adaptive, and tailored library of micro-interventions, which we hope will support more sustainable positive play among a broader cohort of players and enable more timely interventions, reducing escalation into riskier gambling patterns. The empirical evaluation of the POC version of the RG framework will examine a) the feasibility and user acceptance (i.e., To what extent do players engage with the recommended RG tools? Are there differences among different types of RG tools? Are tools perceived positively and useful?), and b) the potential impact of the POC version of the RG framework on the patterns of online gambling behaviors (i.e., Do players' risk levels change or remain equal after exposure to RG micro-interventions?).

Specify the research type and briefly describe the methodology, the types of data to be collected, and the tools used for data collection

POC Evaluation study: A high-fidelity, operational POC version of the RG framework will be systematically evaluated in a naturalistic mixed-methods study with a sequential QUAN-qual study design.

Quantitative data collection: quasi-experimental study lasting over a period of 6 months (i.e., two cycles of 90-day evaluation periods). The POC RG framework will be released to customers of two gambling operators (1 x NL and 1 x UK; RG framework embedded within the operator website). Participant inclusion criteria specify players must be 18+, satisfy all legal requirements to engage in online gambling, having played a minimum of 15 active gambling days (as required to accurately assess gambling risk with the gambling risk assessment algorithm), and consent to share their data for scientific research. No sample size analysis has been conducted due to the study exploratory and preliminary nature (minimum requirement should study implementation not be possible on the entire customer base: a random sample of 30,000 active players stratified by gender and age). No explicit consent will be asked concerning receiving the POC framework on user interface since that is mandated by law; however, all players are given a choice to opt-out at any time and without reason from sharing their anonymized data for scientific research purposes by adjusting

their privacy settings in accordance with European data privacy regulation (GDPR, 2018). Quantitative monitoring and collection during the testing phase will be managed by the gambling operators as also mandated by national regulations.

Feasibility and user acceptance of the RG framework are operationalized into the following outcomes: level of engagement (i.e., duration and frequency of usage, response rate) and user satisfaction with the RG tools delivered during the study period. Potential impact of the POC as a whole will be evaluated by examining the rate of fluctuation in gambling risk as assessed by the gambling risk assessment algorithm over the study period. Individual impact of micro-interventions will not be considered as the focus is on a system of RG tools acting in concert rather than on individual RG measures alone.

Data are recorded automatically in the backend of the gambling operators' websites over the period of 6 months and will include:

1. Player engagement with RG tools will evaluate a user's adoption, interaction with and reaction to RG micro-interventions. Engagement outcomes are based on log data parameters (i.e., web analytics) and will include aggregated usage parameters (i.e., frequency and time spent interacting with a RG tool) and aggregated response rate (i.e., rate of completion of a task when an RG tool includes an immediate call to action) for each RG tool included in the POC. An additional metric of general adoption of the POC system as a whole will be derived from these outcomes, i.e., the percentage of players that interact with any RG tool at all over the study period, which can be useful to compare previous benchmarks of RG tool use within the operator and in the literature.
2. User satisfaction metrics with RG tools assessed via short intercept user satisfaction surveys released live on the operator's website during the study period. These brief surveys will assess the following dimensions of satisfaction, intrusiveness, usability, and utility perception of a given RG tool deployed to the player.
3. Fluctuations in gambling risk data captured by logging the individual risk trajectories over the study period (i.e., player gambling risk scores as assessed daily by the gambling risk assessment algorithm, including resulting coefficients of markers thereof). The gambling risk level at day 1 and day 180 will be additionally used to evaluate the discrete change in risk level at the group level between the start and end of the study.
4. RG tool log data: players will receive variations in RG tool sets (i.e., tools are tailored to risk level assignment, intended proximal outcome and function), and log data on the assignment of RG tools over the study period (i.e., type, amount, timing, randomization probability) will be kept for each player throughout the study period.

At the conclusion of the study period, data will be extracted and anonymized by the operators before being transferred to the researchers for analysis.

Qualitative data collection. Participants will be recruited from the main qualitative sample via purposive sampling. The quantitative sample will be divided into four groups based on the differences in gambling risk between the start and end of the study period. From these four groups, subsamples will be randomly selected, stratified by aggregated level of engagement activity with the RG tools they received. Gender and preferred types of gambling (i.e., the type of gambling games a participant played most frequently and for the longest duration during the study period) will also be taken into account. We aim to interview 10 participants from each of the four risk-assessment trajectory groups, for both sites (UK and NL), for a total of 80 participants. Semi-structured interviews will be conducted. The interviews will take place via videoconferencing (i.e., MTeams), and will be audio recorded for later anonymized verbatim transcription, after which the recordings will be destroyed. The interviews will take approximately 30 minutes to complete.

Specify the [financial and time] resources needed to implement this data management plan

Data management costs are covered internally by collaborating gambling operators (ie, in kind contribution) to manage, anonymize and transfer collected data to EUR research team. Standard data processing and management activities (i.e., data wrangling, analysis and storage) are covered by PhD contract and EUR facilities.

No additional storage space is needed, whereas Open Access publication costs will be covered by VISNU and EUR Open Access publishing agreements with scientific journals.

Preparation: Legal Arrangements and Policy

1. With whom will you need to make legal arrangements?

- With third parties
- For working with multiple research partners
- With research participants

2. List the agreements that you will initiate and with whom will you make them.

Who	Type of agreement
Third-party	• Collaboration agreement (& sponsors): gambling operators (Holland Casino, PlayTech) • Non-disclosure agreement: Playtech • Data sharing agreement: gambling operators (Holland Casino, PlayTech)
Multiple research partners	• Collaboration agreement: UvA • Non-disclosure agreement: UvA
Research participants	• Information brochure and opt-out consent (quantitative study) • Informed consent (qualitative study)
Third Party	• Awarded: funder ICRG

3. List the agreements or other data management policies that you need to uphold but did not initiate. If you are reusing existing data, list the terms of use under which you may re-use them.

Who	Type	Version and Date
MS Teams	Terms of Service	Version [June 15, 2021]
Holland Casino	Terms of Service (Privacy Policy)	Version [October 1, 2021]
PlayTech (VF2011 Limited; U.K., subsidiary of PlayTech Plc including Sun Bingo and Sun Vegas platforms)	Terms of Service (Privacy Policy)	Sun Bingo: Version [May 25, 2018] Sun Vegas: Version [January 7, 2022]
EUR	Internet and ICT facilities policy	V1.0 [2015]
EUR	RDM policy of Erasmus University Rotterdam (EUR)	Version 1.0 [August 14, 2020]
Atlas.ti	Terms of Service	Version [December 15, 2021]
ICRG (if awarded)	Funding agreement	TBC

4. Do you need to obtain ethical approval for your research project?

- Yes, I have submitted my application

5. If you have obtained ethical approval, list the reference number

ETH2122-0810

During research: Collecting and analyzing

6. Specify what data you will be collecting and indicate format, estimated size, and whether this is data that you will be generating or existing data that you will be re-using.

Type	Format	Estimated size	Generate	Re-use
Player engagement metrics data	.csv	15-20GB	Yes	No
User satisfaction survey data	.csv	<10GB	Yes	No
Gambling risk trajectory log data	.csv	<10GB	Yes	No
RG tool deployment log data	.csv	<10GB	Yes	No
Audio-recorded interview + transcripts	.mp3; .docx/.odt	1-5GB	Yes	No

7. Will you be collecting [sensitive] personal data?

- No – My research involves human participants, but I will collect fully anonymous data
- Yes – Personal data [non-sensitive] -->Consult your faculty's Privacy Officer

8. If you collect [sensitive] personal data, how will you protect the privacy of participants when sharing your data?

- I will pseudonymize the data
- I will fully anonymize the data

9. Will you be collecting non-personal sensitive data?

- No

10. Where will you store your data during the project? You can select multiple options.

- EUR SURF Research Drive [for large collaborations]
- Other (please specify where will you store your data in the additional information box).
- EUR OneDrive

Quantitative data collection will initially be handled and stored by the collaborating gambling operators (PlayTech and Holland Casino). At the conclusion of the data collection, quantitative data will be transferred to the EUR. All qualitative data will be collected and stored directly within EUR repositories; EUR Yoda; EUR Google GSuite.

11. Is this other tool / are these other tools supported by the EUR or is it private?

- EUR supported

12. What hardware and software do you use? Select all applicable options.

- Private hardware [e.g. personal laptop, private external hard-drive]
- EUR supported software as found in the software catalog
- EUR supported hardware [e.g. @wEURk laptop, @wEURk workstation]

13. If you use private hardware, software, or freeware, please specify what and for what reason:

Collaborating researchers do not all have EUR issued devices (i.e., laptop), so will be using their own devices.

14. Are regular backups made of your data?

- Yes, I use only EUR supported tools [as listed in Q12], thus regular backups are made automatically

15. Who manages access to the data?

- Researcher responsible for research data management
- Other (please specify who in the additional information box).

When stored by collaborating gambling operators, access to the data is managed under their own data handling policies and GDPR regulation. Once the data is transferred to the EUR, the responsible researchers will control data access.

16. Who will have access to the data (during the project)?

- A third party involved in my research (please specify in the additional information box)
- Only researchers as indicated under 'Administration & Project description'

Third-party access: both collaborating gambling operators (PlayTech & Holland Casino) will have access to their own customer data throughout the project.

17. How are you going to make sure your data will be accessible in case of staff changes, illness, etc?

- There is a clear procedure in place in my research team, department, or faculty
- Other (please specify in the additional information box).

Data stored by the collaborating gambling operators (i.e., pre the data transfer to the EUR) will be under the responsibility of the respective partner. For example, pre the data transfer, the data accessibility will be under the data management policy of the respective industry partners. Once transferred to the EUR, data access will be managed by the lead project researcher Marilisa Boffo (supervisor) and shared with the project manager Harriet Galvin.

Only data in an aggregated format will be published in the Open Science Framework and made accessible to anybody anytime.

18. Have you and your research team agreed on a way to name and order project folders and files?

- Yes - I am working on the documentation

19. Have you and your research team agreed on how to handle versioning of files?

- Yes - I am working on the documentation

After research: Data sharing, re-use

20. What data [and code] will be shared in a research data repository?

- A selection of the data [and code]

21. Please specify why you are unable to share [all] data [and code]

No raw data including point values of individual transactions will be shared open access. Transcripts of qualitative interviews will also not be shared open access to protect participants' confidentiality.

22. List the data [and code] that you plan to share in a research data repository. Also list the information / documentation / metadata that you will include to make the data package self-explanatory and re-usable in the future (for other researchers and yourself)

Data	Format	Size
Anonymized aggregated log data for engagement indicators only	.csv	10-15 GB
Anonymized user satisfaction survey scores	.csv	<10 GB
Anonymized gambling risk scores over time, combined with coefficients of markers thereof and RG tools selection data	.csv	<10 GB
Qualitative interview script	.pdf	<1 GB
Readme text file (general description of the data, incl. date of collection, selection procedure of participants, general guide to RG tools used to collect the data, etc.)	.txt	<1 GB
Qualitative results (codes and themes)	.txt	<1 GB

23. In which repository will you place the [meta]data and/or code that are associated with your paper?

- EUR Data Repository (EDR)

Open Science Framework (OSF)

Data pre-data transfer to the EUR will be stored by the collaborating gambling operators.

24. What metadata standard will you use to document your research?

- None
- DCMI [Dublin Core Metadata Initiative] (Note: Default within the EUR Data Repository)

25. Will you place any restrictions on re-using of data?

- Yes, permanent restrictions – Restricted access

Based on the conditions of the collaboration agreement between the EUR and the collaborating gambling operators, data re-use will be limited to specific conditions agreed among all collaborating partners.

27. Please specify the conditions under which data with restricted access may be re-used:

Openly shared data may be re-used solely to replicate the study analyses or to conduct meta-analytical analyses of intervention effects. No commercial use of the data will be allowed.

28. Under what license will you make your data available for re-use?

- Other (please specify in the additional information box).

[CC BY-NC-SA](#)

29. Please specify which license

CC BY-NC-SA 4.0

After research: Archiving

30. You may be obliged to destroy some data before archiving. Do any of such obligations apply to you?

- Yes - Privacy law [e.g. personal data of participants]

Qualitative study: Interviews will be audio recorded for later anonymized verbatim transcription, after which the recordings will be destroyed. Transcripts of qualitative interviews will not be shared open access to protect participants' confidentiality. Further, participant anonymity will be preserved by removing any possible personal and/or sensitive information from the verbatim transcripts of the interviews.

31. List the data you will be archiving. These data constitute your archival package.

Data	Format	Size
Collaboration Agreement	.pdf	<1 GB
Non-Disclosure agreement	.pdf	<1 GB
Data sharing agreements (1 per collaborator)	.pdf	<1 GB
Consortium Agreement (agreement between the collaborating university entities only; UvA & EUR)	.pdf	<1 GB
ICRG grant application & related material.	.pdf	<1 GB
Ethical review application & approval document	.pdf	<1 GB
Data management plan	.pdf	<1 GB
Responsible gambling framework (incl. conceptual framework; literature review, tool matrix, references guide)	.docx, .csv,	1-5 GB
Technical production documentation (e.g., user journey analyses log data output guidelines)	.docx	1-5 GB
Raw anonymized quantitative data (received from industry partners)	.csv	40-50 GB
Processed quantitative data (conducted by EUR - aggregated for open access sharing)	.csv	20-30 GB
Qualitative informed consent forms (signed)	.pdf.	<1 GB
Processed qualitative data (transcripts)	.docx	<1 GB

32. Where will you be archiving your data?

- Archive of EUR Library [retention period min. 10 years]